

Frederick Joseph Boehm, III

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Academic Appointment

South Dakota State University

Assistant Professor, Department of Mathematics and Statistics

Brookings, South Dakota, USA

08/2024 to present

Education and Training

University of Michigan

Postdoctoral Research Fellow, Department of Internal Medicine ([NIH T32 HL007853](#))

Postdoctoral Research Fellow, Department of Biostatistics

Ann Arbor, Michigan, USA

07/2023 to 06/2024

06/2021 to 05/2023

University of Massachusetts Medical School

Postdoctoral Research Fellow, Department of Microbiology & Physiological Systems

Worcester, Massachusetts, USA

09/2019 to 05/2021

University of Wisconsin-Madison

Postdoctoral Research Fellow, Department of Biostatistics and Medical Informatics

Ph.D., Department of Statistics

Postdoctoral Research Fellow ([NIH T32 ES007015](#)), Molecular and Environmental Toxicology Center

Assistant Researcher, [Value-Added Research Center](#)

Madison, Wisconsin, USA

06/2019 to 08/2019

09/2011 to 05/2019

02/2012 to 01/2014

01/2011 to 07/2011

University of Washington

Postdoctoral Research Fellow, Department of Biostatistics

Seattle, Washington, USA

11/2007 to 09/2009

University of Wisconsin-Madison

M.D., School of Medicine and Public Health

M.S., Department of Population Health Sciences

Graduate Project Assistant, Department of Population Health Sciences

Madison, Wisconsin, USA

08/2003 to 08/2007

01/2003 to 08/2007

01/2003 to 05/2003

La Clinica de los Campesinos

Spanish Language Medical Interpreter

Wautoma, Wisconsin, USA

06/2002 to 08/2002

University of Wisconsin-Madison

B.S., with Honors in Chemistry, Mathematics, and Biochemistry

Madison, Wisconsin, USA

09/1996 to 05/2001

Written Works

UNPUBLISHED MANUSCRIPTS

1. Adebawale Awosika-Olumo, **Frederick J. Boehm**, RinoRinojj Gautam, and Usha Sambamoorthi (submitted). Predicting Human Papillomavirus Vaccination Uptake Among U.S. adults Using BRFSS 2023: A Cross-Sectional Machine Learning Study.
2. Alexander Katz, Cameron B. Pinnock, Ayse Bilge Ozel, **Frederick J. Boehm**, Andrea Murad, Min-Lee Yang, Kristina L. Hunker, Natalia Fendrikova Mahlay, Dawn M. Coleman, Matthew Zawistowski, Jeffrey Olin, James Froehlich, Jason Kovacic, Heather Gornik, James C. Stanley, Xiang Zhou, and Santhi K. Ganesh (submitted). Rare coding variation in cell-matrix adhesion and trimer collagen genes influences disease susceptibility and severity in fibromuscular dysplasia.
3. Peng Zhou, Ji Hoon Park, and **Frederick J. Boehm** (in preparation). ShinyMR: A web platform to promote rigor and reproducibility in Mendelian randomization studies.
4. Emma G. Brookman and **Frederick J. Boehm** (in preparation). Preferential sampling of environmental monitoring networks method comparison: A Bayesian MCMC framework and INLA.
5. Shree Nyaupane and **Frederick J. Boehm** (in preparation). Rare variant association testing with XGBoost weights.
6. Christine Larson and **Frederick J. Boehm** (in preparation). Visible Thinking in Calculus.

JOURNAL ARTICLES

7. Christina G. Hutten, **Frederick J. Boehm**, Jennifer A. Smith, Brian W. Spitzer, Sylvia Wassertheil-Smoller, Carmen R. Isasi, Jianwen Cai, Jonathan T. Unkart, Jiehuan Sun, Victoria Persky, Martha L. Daviglus, Tamar Sofer, and Maria Argos (2025). Differential

- performance of polygenic risk scores for heart disease in Hispanic/Latino subgroups: Findings of the Hispanic Community Health Study/Study of Latinos. *Human Genetics and Genomics Advances* 6(4):100486. <https://doi.org/10.1016/j.xhgg.2025.100486>.
8. Sherry L. Kurtz, Richard E. Baker, **Frederick J. Boehm**, Chelsea C. Lehman, Lara R. Mittereder, Hamda Khan, Amy P. Rossi, Daniel M. Gatti, Gillian Beamer, Christopher M. Sasseti, and Karen L. Elkins (2024). Multiple genetic loci influence vaccine-induced protection against Mycobacterium tuberculosis in genetically diverse mice. *PLOS Pathogens* 20(3):e1012069. <https://doi.org/10.1371/journal.ppat.1012069>.
 9. **Frederick J Boehm** and Xiang Zhou (2022). Statistical methods for Mendelian randomization in genome-wide association studies: A review. *Computational and Structural Biotechnology Journal* 20:2338-2351. <https://doi.org/10.1016/j.csbj.2022.05.015>.
 10. Clare M Smith, Richard E Baker, Megan K Proulx, Bibhuti B Mishra, Jarukit E Long, Sae Woong Park, Ha-Na Lee, Michael C Kiritsy, Michelle M Bellerose, Andrew J Olive, and others (2022). Host-pathogen genetic interactions underlie tuberculosis susceptibility in genetically diverse mice. *Elife* 11:e74419. <https://doi.org/10.7554/eLife.74419>.
 11. **Frederick J Boehm** and Bret M Hanlon (2021). What is happening on Twitter? A framework for student research projects with tweets. *Journal of Statistics and Data Science Education* 29:S95-S102. <https://doi.org/10.1080/10691898.2020.1848486>.
 12. Jorge L Rodriguez-Gil, Dawn E Watkins-Chow, Laura L Baxter, Gene Elliot, Ursula L Harper, Stephen M Wincovitch, Julia C Wedel, Arturo A Incao, Mylene Huebecker, **Frederick J Boehm**, and others (2020). Genetic background modifies phenotypic severity and longevity in a mouse model of Niemann-Pick disease type C1. *Disease Models & Mechanisms* 13(3):dmm.042614. <https://doi.org/10.1242/dmm.042614>.
 13. **Frederick J. Boehm**, Elissa J. Chesler, Brian S. Yandell, and Karl W. Broman (2019). Testing pleiotropy vs. separate QTL in multiparental populations. *G3* 9(7):2317. <https://doi.org/10.1534/g3.119.400098>.
 14. Julia H Kemis, Vanessa Linke, Kelsey L Barrett, **Frederick J Boehm**, Lindsay L Traeger, Mark P Keller, Mary E Rabaglia, Kathryn L Schueler, Donald S Stapleton, Daniel M Gatti, and others (2019). Genetic determinants of gut microbiota composition and bile acid profiles in mice. *PLoS genetics* 15(8):e1008073. <https://doi.org/10.1371/journal.pgen.1008073>.
 15. **Frederick Boehm**, Brian Yandell, and Karl W Broman (2019). qtl2pleio: Testing pleiotropy vs. separate QTL in multiparental populations. *Journal of Open Source Software* 4(38):1435. <https://doi.org/10.21105/joss.01435>.
 16. Chris Wells, Dhavan V. Shah, Jon C. Pevehouse, JungHwan Yang, Ayellet Pelled, **Frederick Boehm**, Josephine Lukito, Shreenita Ghosh, and Jessica L. Schmidt (2016). How Trump Drove Coverage to the Nomination: Hybrid Media Campaigning. *Political Communication* 33(4):669-676. <https://doi.org/10.1080/10584609.2016.1224416>.
 17. Eric C Tauchman, **Frederick J Boehm**, and Jennifer G DeLuca (2015). Stable kinetochore-microtubule attachment is sufficient to silence the spindle assembly checkpoint in human cells. *Nature communications* 6(1):10036. <https://doi.org/10.1038/ncomms10036>.
 18. William F Dove, Alexandra Shedlovsky, Linda Clipson, James M Amos-Landgraf, Richard B Halberg, Kathleen J Krentz, **Frederick J Boehm**, Michael A Newton, David J Adams, and Thomas M Keane (2014). A strategy to identify dominant point mutant modifiers of a quantitative trait. *G3: Genes| Genomes| Genetics* 4(6):1113-1121. <https://doi.org/10.1534/g3.114.010595>.
 19. James C Costello, Laura M Heiser, Elisabeth Georgii, Mehmet Gönen, Michael P Menden, Nicholas J Wang, Mukesh Bansal, Petteri Hintsanen, Suleiman A Khan, John-Patrick Mpindi, and others (2014). A community effort to assess and improve drug sensitivity prediction algorithms. *Nature biotechnology* 32(12):1202-1212. <https://doi.org/10.1038/nbt.2877>.
 20. Cathy C Laurie, Kimberly F Doheny, Daniel B Mirel, Elizabeth W Pugh, Laura J Bierut, Tushar Bhangale, **Frederick Boehm**, Neil E Caporaso, Marilyn C Cornelis, Howard J Edenberg, and others (2010). Quality control and quality assurance in genotypic data for genome-wide association studies. *Genetic epidemiology* 34(6):591-602. <https://doi.org/10.1002/gepi.20516>.
 21. MC Smith Fawzi, W Lambert, **F Boehm**, JL Finkelstein, JM Singler, F Léandre, P Nevil, D Bertrand, MS Claude, J Bertrand, and others (2010). Economic Risk Factors for HIV Infection Among Women in Rural Haiti: Implications for HIV Prevention Policies and Programs in Resource-Poor Settings. *Journal of Women's Health* 19(5):885-892. <https://doi.org/10.1089/jwh.2008.1334>.
 22. Elizabeth E Marchani, Yanming Di, Yoonha Choi, Charles Cheung, Ming Su, **Frederick Boehm**, Elizabeth A Thompson, and Ellen M Wijsman (2009). Contrasting identity-by-descent estimators, association studies, and linkage analyses using the Framingham Heart Study data. *BMC proceedings* 3(Suppl 7):S102. <https://doi.org/10.1186/1753-6561-3-S7-S102>.
 23. Jason E Gestwicki, Laura E Strong, Christopher W Cairo, **Frederick J Boehm**, and Laura L Kiessling (2002). Cell aggregation by scaffolded receptor clusters. *Chemistry & biology* 9(2):163-169. [https://doi.org/10.1016/S1074-5521\(02\)00102-3](https://doi.org/10.1016/S1074-5521(02)00102-3).

EDITED BOOK

24. A Vishnu Moorthy, Bryan Neil Becker, **Frederick J Boehm III**, and Arjang Djamali, editors (2009). *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.

BOOK CHAPTERS

25. Ayellet Pelled, Josephine Lukito, **Frederick Boehm**, JungHwan Yang, and Dhavan Shah (2018). Little Marco', Lyin' Ted', Crooked Hillary', and the Biased' Media: How Trump Used Twitter to Attack and Organize. In Natalie Stroud and Shannon McGregor, editors *Digital Discussions: How Big Data Informs Political Communication*. Routledge.

26. Vishnu Moorthy and **Frederick Boehm** (2009). Urinalysis and an Approach to Kidney Diseases. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.
27. Kelly Traeger, **Frederick Boehm**, and Arjang Djamali (2009). Acid-Base Homeostasis and Metabolic Alkalosis. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.
28. **Frederick Boehm**, Kelly Traeger, and Arjang Djamali (2009). Metabolic Acidosis and Approach to Acid-Base Disorders. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.

SOFTWARE

29. **Frederick Boehm** (2018-06-04). qtl2pleio R package. <https://github.com/fboehm/qtl2pleio>.
30. **Frederick Boehm** (2018-03-15). gemma2 R package. <https://github.com/fboehm/gemma2>.

CONFERENCE PROCEEDINGS

31. J Mark, L Wesler, C Oswald, F Léandre, P Nevil, D Bertrand, J Bertrand, **F Boehm**, and MC Smith Fawzi (2011). P2-S9. 07 Economic risk factors for syphilis infection among pregnant women in rural Haiti. In *Sexually Transmitted Infections*.
32. JE Gestwicki, CW Cairo, **FJ Boehm**, and LL Kiessling (2001). Mechanistic origins of the valency effect: Multivalent ligand binding studied by surface plasmon resonance and isothermal titration calorimetry.. In *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY*.
33. E Fasella, **FJ Boehm**, and LL Kiessling (2000). Synthesis of galactosyl-alpha (1-3)-galactosyl epitopes for use in immunological studies.. In *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY*.

PH.D. THESIS

34. **Frederick J. Boehm III** (2019). Testing pleiotropy vs. separate QTL in multiparental populations.

OTHER PUBLICATIONS

35. Catherine DeRose, Carrie Roy, and **Fred Boehm** (2014). Victorian eyes: Literary, statistical, and artistic perspectives on Victorian novels – and Dickens’s unfinished murder mystery. *Significance*.
36. **Fred Boehm** (1997). The lone male wolf of the Sandhill Wildlife Area. *International Wolf*.

Research Funding

FUNDED

Conference: SDSU Data Science Symposium December 2024
Principal investigator · National Science Foundation · Total direct costs: \$7,502 · Award No. 2515675

UNDER REVIEW

Targeting POLR1A-dependent ribosome biogenesis as a new therapeutic approach for advanced liver cancer June 2026
Co-investigator (PI: Bilal Hafeez) · National Institutes of Health (R15) · Total direct costs: \$375,000

Bayesian Hierarchical Modeling for Multi-trait, Multi-ancestry Polygenic Scores June 2026
Principal investigator · National Institutes of Health (R15) · Total direct costs: \$361,945

Collaborative Research: FEC: StaTIC: Structure- and Data-Driven Discovery of Molecular Glues for Transient Transcription-Factor Complexes January 2026
Co-investigator (PI: Suvobrata Chakravarty) · National Science Foundation (EPSCoR FEC) · Total direct costs: \$3,130,401

Expanding and Analyzing Genetic Diversity Across Bison Herds November 2025
Co-investigator (PI: Jose Gonzalez) · USDA NIFA · Total direct costs: \$454,970

NOT FUNDED

Measuring the accuracy of multi-trait, multi-ancestry polygenic scores March 2026
Principal investigator · South Dakota Board of Regents (Competitive Research Grant) · Total direct costs: \$86,959

Measuring the impact of linkage disequilibrium panels on polygenic score accuracy November 2025
Principal investigator · Healthy Americas Research Consortium · Total direct costs: \$10,000

Measuring the Impact of LD Reference Panel Choice on PGS Accuracy in Admixed Populations November 2025
Principal investigator · BioSNTRii · Total direct costs: \$75,000

Teaching Experience

FORMAL COURSES

Instructor, Bioinformatics, South Dakota State University Fall 2026
Guest lecturer, Career Seminar, South Dakota State University Spring 2025
Instructor, Senior Capstone I, South Dakota State University Spring 2025
Instructor, Statistical methods II, South Dakota State University Spring 2025, Spring 2026
Instructor, Time series analysis, South Dakota State University Fall 2024, Fall 2025, Fall 2026
Co-facilitator, Inclusive teaching for educating a diverse nation, University of Wisconsin-Madison Delta Program & CIRTl Network Spring 2017, Fall 2017
Guest lecturer on topic models & Teaching assistant, Special Topics: Data Science, University of Wisconsin-Madison Fall 2016
Instructor, Introductory statistics, University of Wisconsin-Madison Fall 2016, Spring 2016, Fall 2015, Summer 2015, Spring 2015, Fall 2014, Spring 2014
Teaching assistant, Introduction to Biostatistics, University of Washington Fall 2009, Winter 2010
Teaching assistant, Introductory organic chemistry, University of California, Berkeley Fall 2001

SHORT COURSES & WORKSHOPS

Instructor, Engaging students in the teaching of statistics, CIRTl Network March 2018, July 2020
Instructor, [Data Carpentry Workshop](#), University of Wisconsin-Madison, Missouri S&T, George Mason University June 2017, September 2018, January 2019
Instructor, Data analysis with topic models for communications researchers, University of Wisconsin-Madison July 2016, October 2016
Instructor, Topic modeling for non-statisticians, University of Wisconsin-Madison October 2013
Instructor, Statistical computing with R, University of Wisconsin-Madison June 2012, June 2013

TRAINING IN TEACHING

Postdoctoral Short Course on College Teaching in STEM, University of Michigan Center for Research on Learning and Teaching Spring 2024
Expeditions in Learning: Exploring How Students Learn with Writing Across the Curriculum, University of Wisconsin-Madison Delta Program Spring 2016
Diversity in the College Classroom, University of Wisconsin-Madison Delta Program Fall 2015
Research Mentor Training, University of Wisconsin-Madison Delta Program Summer 2016 & Summer 2012
An Introduction to Evidence-Based Undergraduate STEM Teaching, CIRTl Network Fall 2014
The College Classroom, University of Wisconsin-Madison Delta Program Summer 2014
Instructional Materials Development: Integrating Statistics into STEM courses, University of Wisconsin-Madison Delta Program Spring 2014
Teaching Statistics in the Classroom, University of Wisconsin-Madison Delta Program Spring 2013

Mentoring Experience

Research mentor to Robert Turner, undergraduate researcher May 2015 to May 2016
Project: Topic Modeling of Tweets to Detect Social Media Events
Next position: Software Engineer at Epic Systems
Research mentor to Jinyu Xia, undergraduate researcher September 2015 to August 2016
Project: Tweet Sentiments and Stock Market Index Trends
Next position: Statistics graduate student at University of Washington
Research mentor to Ji Hoon Park, undergraduate researcher November 2024 to May 2025
Project: Building a Shiny App for Mendelian Randomization Analysis
Next position: Graduate student researcher
Research mentor to Emma Brookman, graduate researcher June 2025 to present
Project: Causal Inference Methods in Environmental Epidemiology
Research mentor to Etornam Kunu, graduate researcher September 2025 to present
Project: Polygenic Scores in Diverse Populations
Research mentor to Shree Nyaupane, graduate researcher September 2025 to present
Project: Rare Variants in Complex Traits and Diseases
Research mentor to Henrietta Kotey, graduate researcher September 2025 to May 2026

Project: Multivariable Mendelian Randomization Methods
Research mentor to Peng Zhou, Gradaute researcher June 2026 to present
 Project: ShinyMR: A web application to promote rigor and reproducibility in Mendelian randomization studies

Leadership Training

Participant, University of Michigan You3 Leadership Training Program Spring 2024
 Participant, University of Michigan Designing Your Life Course Fall 2022
[Participant, Leadership and Management in Action Program](#) Fall 2021
[Committee Member, Genetics Society of America Early Career Researchers Leadership Program](#) February 2020 to February 2022
 Participant, American Statistical Association Workshop on Communicating with the Media and the Public November 2016

Awards & Honors

Conference Travel Award, University of Michigan Postdoc Association 2024
 Best Poster Award, International Mammalian Genome Conference 2018
 Conference Travel Scholarship, International Mammalian Genome Conference 2018
 Integrating Research Ethics and Scholarship Fellowship, University of Wisconsin-Madison 2013
 New Arts Venture Challenge Award (for Victorian Eyes art exhibition), University of Wisconsin-Madison 2013
 Medical Student Research Training Fellowship, Howard Hughes Medical Institute 2006
 Medical Student Summer Research Stipend, University of Wisconsin-Madison Department of Family Medicine 2004
 Graduate Research Fellowship, National Science Foundation 2001
 Meyerhoff Undergraduate Excellence Award, University of Wisconsin-Madison 2001
 University Book Store Award for Outstanding Undergraduate Thesis, University of Wisconsin-Madison 2001
 Edward Panek Memorial Scholarship, University of Wisconsin-Madison Department of Chemistry 2000
 Elected as Member, The Phi Beta Kappa Society 1999
 Wisconsin Idea Fellowship (for the Salvation Army Learning Center), University of Wisconsin-Madison 1999
 Dean's List, University of Wisconsin-Madison 1998 to 2001
 Summer Research Scholarship, University of Wisconsin-Madison Department of Chemistry 1998
 College of Letters & Science Honors Program Member, University of Wisconsin-Madison 1996 to 2001
 Wisconsin Academic Excellence Scholarship, University of Wisconsin-Madison 1996 to 2000

Presentations

SLIDE PRESENTATIONS

Exploring Human Biology with Genome-Wide Association Studies December 03, 2025
 Sanford Research Seminar Series
 Sioux Falls, South Dakota, United States of America
Exploring Human Biology with Genome-Wide Association Studies October 31, 2025
 South Dakota State University Corothers Lecture
 Brookings, South Dakota, United States of America
My Research Goals September 15, 2025
 South Dakota State University New Faculty Forum
 Brookings, South Dakota, United States of America
Introduction to our Research Team August 19, 2025
 South Dakota State University Graduate Student Orientation
 Brookings, South Dakota, United States of America
shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies August 04, 2025
 Joint Statistical Meetings 2025
 Nashville, Tennessee, United States of America
shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies July 31, 2025
 Institute for Mathematical Statistics New Researchers Conference 2025
 Nashville, Tennessee, United States of America
shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies May 21, 2025
 STATGEN2025 Conference
 Minneapolis, Minnesota, United States of America
A Shiny App to Promote Rigor and Reproducibility in Mendelian Randomization Studies February 07, 2025

South Dakota State University Data Science Symposium Brookings, South Dakota, United States of America	July 30, 2018
<i>Distinguishing pleiotropy from separate QTL in multiparental populations</i> Joint Statistical Meetings Vancouver, British Columbia, Canada	June 20, 2018
<i>Distinguishing pleiotropy from separate QTL in multiparental populations</i> Complex Trait Community Meeting Glasgow, Scotland, United Kingdom	June 30, 2016
<i>Event Detection with Social Media Data</i> UseR! 2016 Meeting Stanford, California, United States of America	November 28, 2012
<i>Post-GWAS characterization of breast cancer-associated SNPs</i> University of Wisconsin-Madison Department of Statistics Student Seminar Madison, Wisconsin, United States of America	September 20, 2012
<i>Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs</i> University of Wisconsin-Madison Molecular and Environmental Toxicology Center Seminar Madison, Wisconsin, United States of America	
POSTER PRESENTATIONS	
<i>shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies</i> Institute for Mathematical Statistics New Researchers Conference 2025 Nashville, Tennessee, United States of America	July 31, 2025
<i>shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies</i> Human Genetics and Genomics Gordon Conference Portland, Maine, United States of America	July 07, 2025
<i>Testing pleiotropy vs. separate QTL in multiparental populations</i> International Mammalian Genome Conference Rio Mar, Puerto Rico, United States of America	November 11, 2018
<i>Testing pleiotropy vs. separate QTL in multiparental populations</i> University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Training Program Retreat Madison, Wisconsin, United States of America	October 26, 2018
<i>Event Detection with Topic Modeling of Tweets</i> Society for Political Methodology's Annual Summer Conference (PolMeth XXXIV) Madison, Wisconsin, United States of America	July 13, 2017
<i>Pleiotropy vs. Close Linkage in the Diversity Outbred Mouse Population</i> The Jackson Laboratory Systems Genetics Short Course Bar Harbor, Maine, United States of America	October 22, 2016
<i>Pleiotropy vs. Close Linkage in the Diversity Outbred Mouse Population</i> University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Training Program Retreat Madison, Wisconsin, United States of America	October 07, 2016
<i>Data Analysis with Topic Models for Communications Researchers</i> Association for Education in Journalism and Mass Communication Annual Meeting Minneapolis, Minnesota, United States of America	August 06, 2016
<i>Event Detection with Topic Modeling of Tweets</i> University of Wisconsin-Madison WARF Discovery Challenge Symposium Madison, Wisconsin, United States of America	April 06, 2016
<i>Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs</i> University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Retreat Madison, Wisconsin, United States of America	October 12, 2012
<i>Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs</i> University of Wisconsin-Madison Molecular and Environmental Toxicology Center Retreat Madison, Wisconsin, United States of America	May 22, 2012

Service

UNIVERSITY SERVICE

Thesis Committee Member for Wilberforce Arthur, South Dakota State University May 2026
South Dakota State University College of Engineering Day Participant, South Dakota State University College of Engineering September 2024, October 2025
Outreach & Recruitment Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2025 to present
Seminar Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2025 to present
Co-organizer, South Dakota State University Data Science Symposium February 2025
Statistics Curriculum Revision Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to present
Data Science Curriculum Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to May 2025
Senior Capstone Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to May 2025
Co-coordinator, University of Wisconsin-Madison Computational Methods in Mass Communications Research Group September 2016 to December 2016
Coordinator, University of Wisconsin-Madison WISCIENCE Graduate Student Book Group June 2015 to May 2016
Organizer, University of Wisconsin-Madison Data, Humanities, and Art (DH + A) Symposium September 2013 to November 2013
Moderator, University of Wisconsin-Madison Undergraduate Research Symposium April 2013, April 2012
Member, University of Wisconsin-Madison Graduate Student Peer Mentor Award Review Committee April 2013
Member, University of Wisconsin-Madison Graduate School Vilas Travel Grant Review Committee October 2012 to November 2012
Member, University of Wisconsin-Madison Statistics Department Social Committee January 2012 to May 2014

COMMUNITY SERVICE

Coach, Westborough Mathcounts Team September 2020 to April 2022
[Penpal](#), Letters to a Pre-Scientist September 2018 to May 2024
Coach, Madison Science Olympiad Team November 2013 to March 2014
Assistant coach, Madison Mathcounts Team September 2012 to May 2013
[Volunteer](#), MEDiC September 2003 to May 2005
Founding director, Salvation Army Learning Center September 1998 to May 2000
Cub Scouts Den Leader, Boy Scouts of America September 1998 to June 2000

PROFESSIONAL SERVICE

Peer reviewer for abstracts submitted to the ASHG Annual Meeting, American Society for Human Genetics June 2026
Co-Secretary, American Statistical Association JEDI Outreach Group January 2026 to present
Session Chair, Joint Statistical Meetings August 2025
Session Chair, ENAR Annual Meeting March 2025
Ad hoc Reviewer, National Science Foundation December 2024
DNA Day Essay Judge, American Society for Human Genetics March 2024
Participant, ENAR Fostering Diversity in Biostatistics Virtual Workshop March 2021
[Associate Editor](#), Journal of Open Source Software June 2020 to present
Peer Reviewer, Journal of Open Source Software, European Journal of Epidemiology, The R Journal, ROpenSci, Genetics, G3, PLoS Computational Biology, BMC Cancer, Journal of Statistics and Data Science Education January 2019 to present
[Trainee](#), Genetics Society of America Peer Review Training Program March 2018 to March 2020
[Instructor](#), The Carpentries May 2017 to present
Participant, Software Carpentry Instructor Training October 2016